

Designing Aula Conectada Through a Gender-Inclusive Evidence Lens

Investment case · Bolivia

Hernando Grueso, PhD

UNICEF

hgrueso@unicef.org

June 23, 2026

1 · Diagnostic — gaps in education

Target population and data sources

We draw on **two independent INE surveys**: **EH 2024** (education, access) and **EDSA 2023** (the home environment — violence, discipline). At the overlapping age band (15–19) they agree closely, so the two can be read together with confidence.

Indicator — adolescent girls 15–19, weighted	EH 2024	EDSA 2023
Currently attending school	86.0%	84.8%
Mean years of schooling	9.95	10.0
% rural	25.5%	26.3%
% indigenous	17.0%	16.2%
Unweighted n (15–19)	3,530	2,224

In absolute terms, Bolivia has roughly **1.19 million adolescent girls aged 10–19** (EH 2024, weighted). Attendance is near-universal at 10–14 (97.9%) but falls to **85.9% by 15–19** — so about **81,000 girls aged 15–19 are out of school**, concentrated at the **senior-secondary transition** (the 10–19 average is 92.1%).

Key takeaways



Bolivian girls are **already in school** (~92% enrolment 10–19, no gender gap in access) — but attendance falls to **86% by 15–19**. The binding problems lie in what marginalised girls learn, whether they stay through the senior-secondary transition, and the home environment they learn from.

What the diagnosis shows

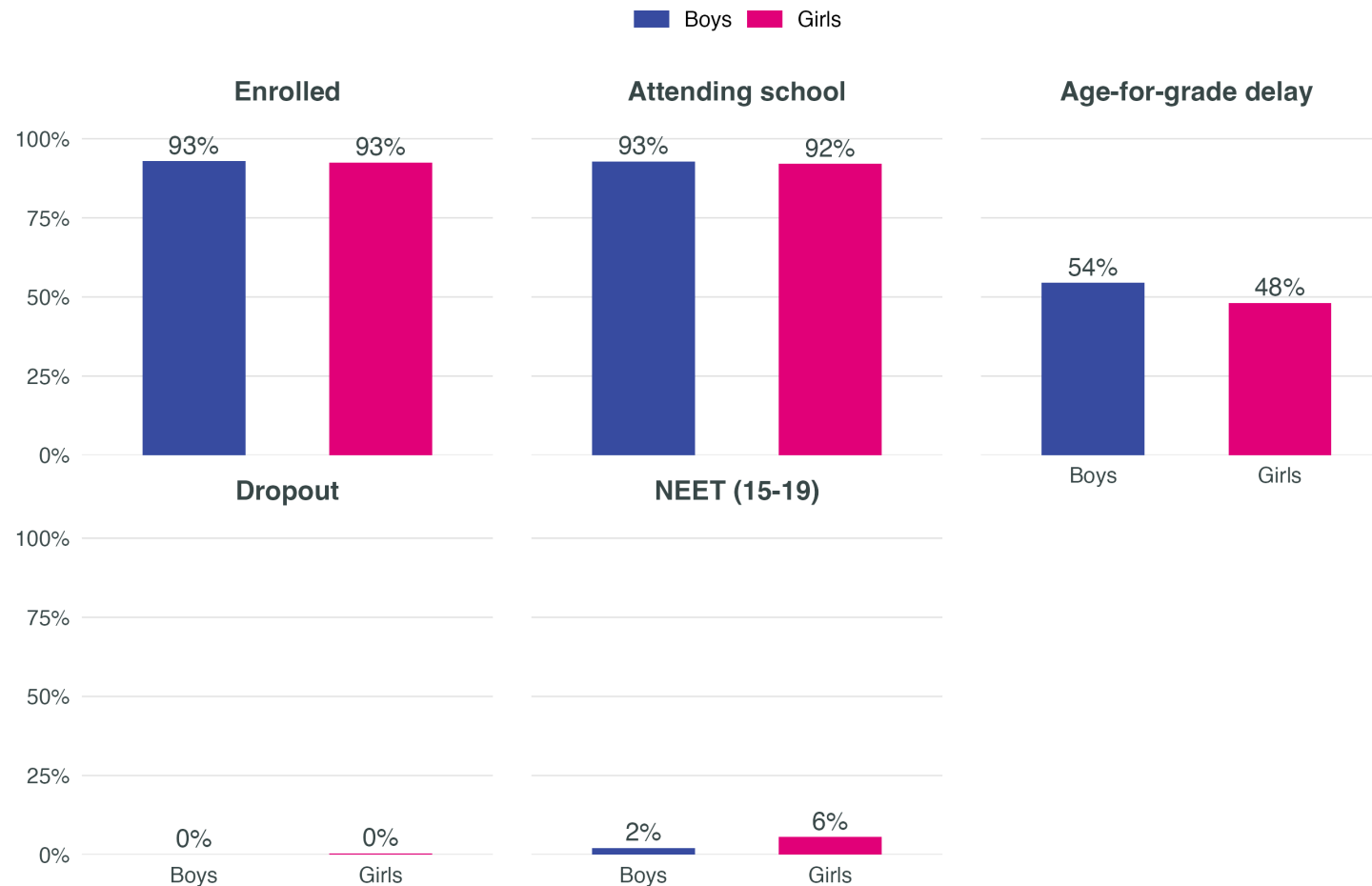
- The gap is **not access** — it is learning quality and **staying in school past 15**, concentrated among **rural, indigenous and poor girls**.
- **Violence at home is part of the story**. Violent discipline is concentrated in **rural and indigenous** homes (rural 52% vs 32% urban; indigenous 46% vs 35%), and girls in harsh-discipline homes have **~2 fewer years of schooling**.
- **Early pregnancy** is the single strongest schooling correlate (**-2.9 years**) — a second, distinct vulnerability for adolescent girls.

What this implies for Aula Conectada

- Fund the **learning layer** — gender-responsive pedagogy and teacher training (Evans & Yuan, 73+ RCTs) — not hardware.
- Add an **empowerment layer** (ELA-style) — the retention answer, addressing drop-out drivers including **early pregnancy**.
- Address the **home environment**: a low-cost **parenting** component for the violence pathway.
- Target the **most vulnerable**: rural and indigenous girls.

Gender gaps in education

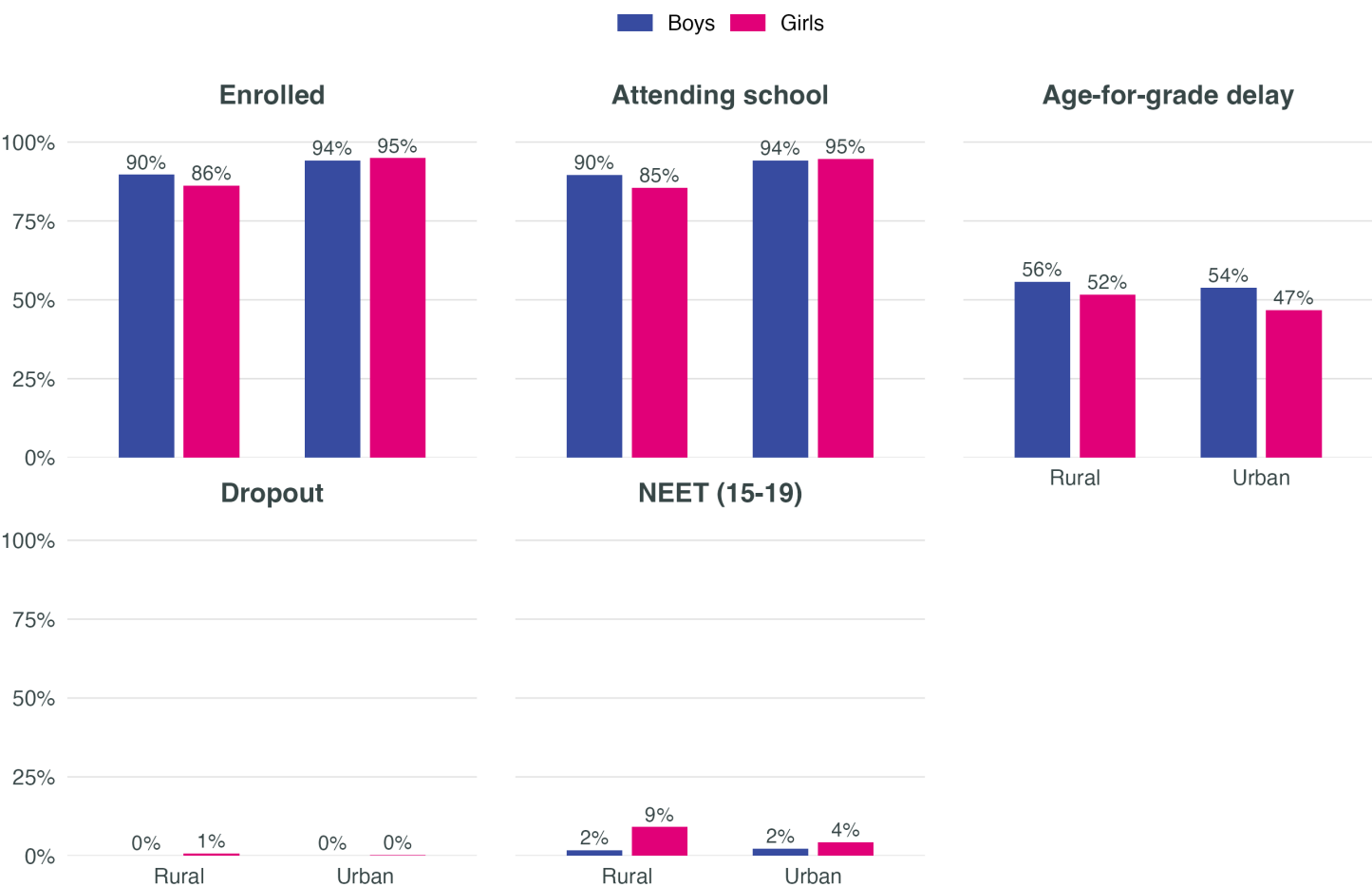
Adolescents 10–19 in Bolivia



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by area of residence

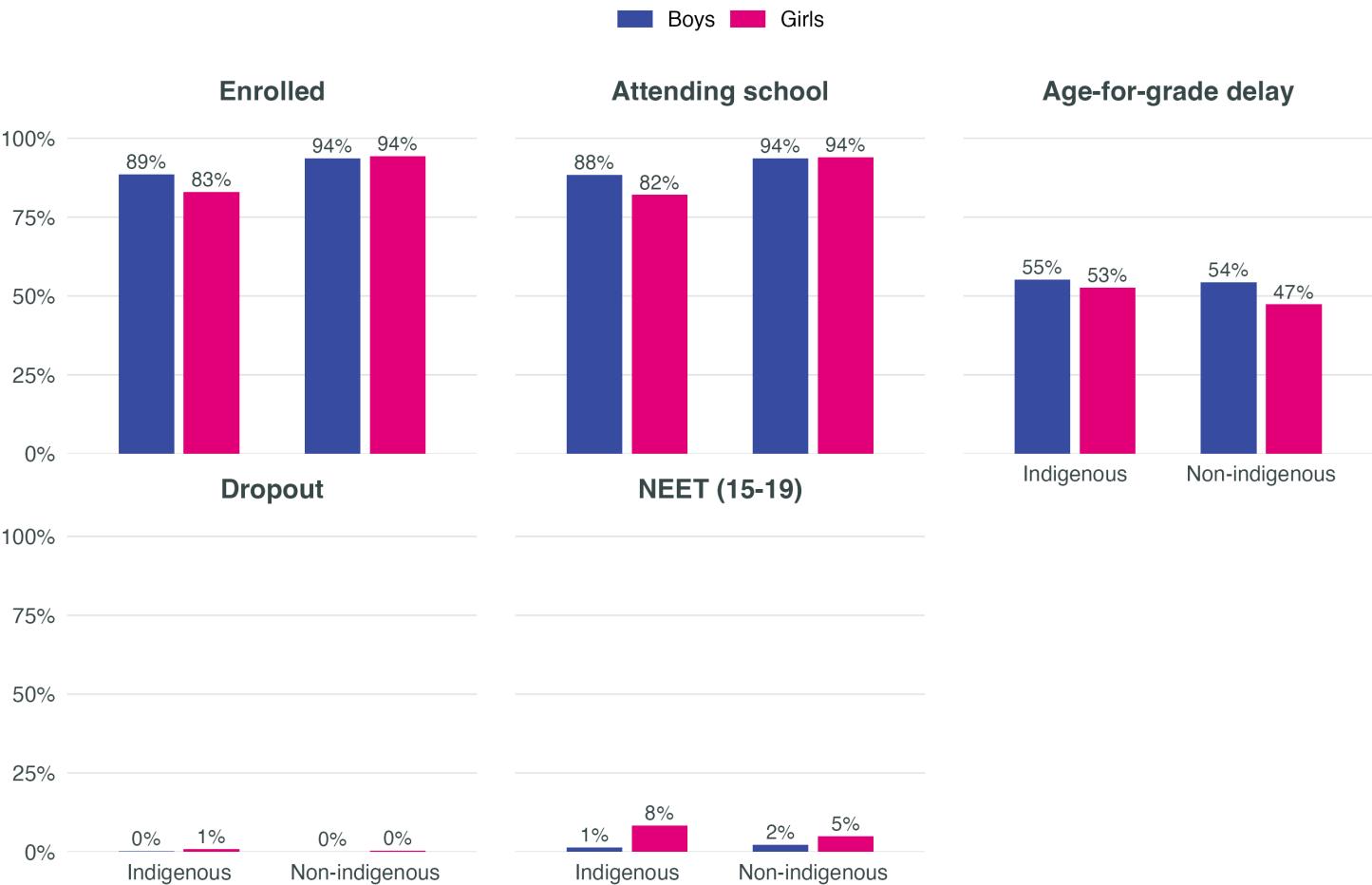
Adolescents 10–19, by sex and rural/urban area



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

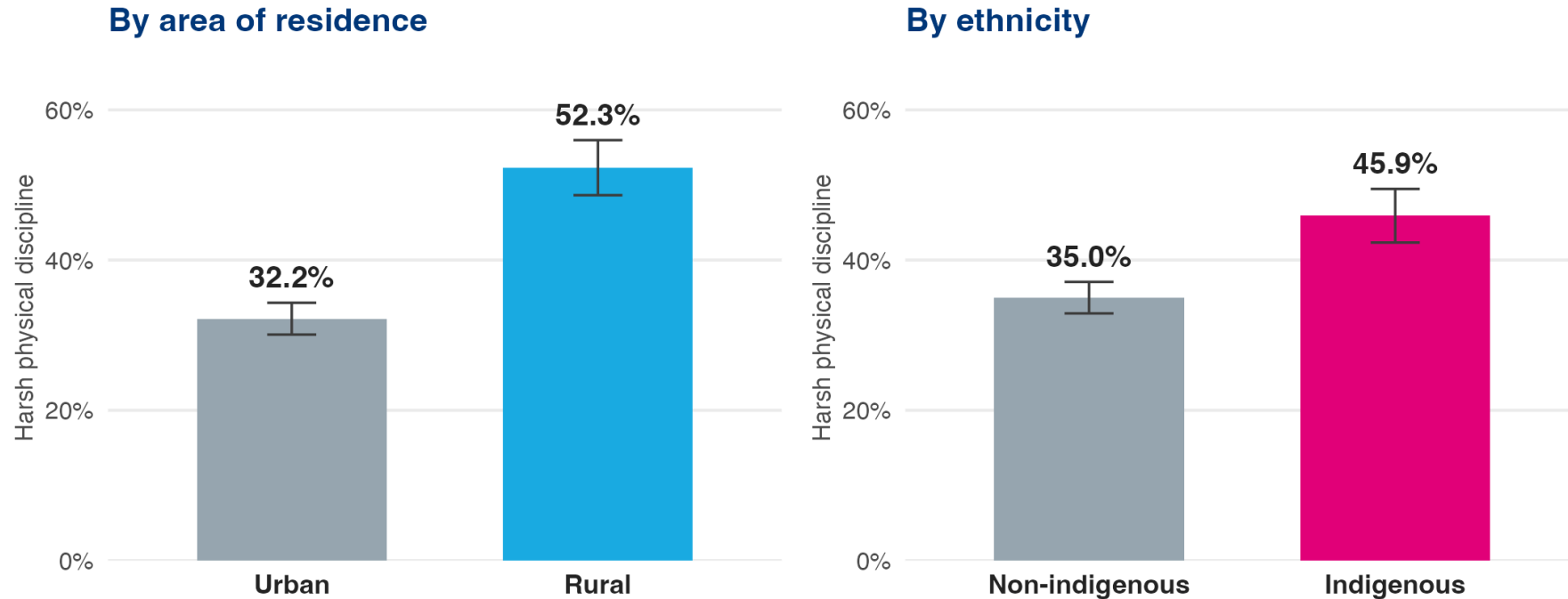
Education outcomes by indigenous status

Adolescents 10–19, by sex



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

The home environment: violent discipline is higher in rural and indigenous homes



EDSA 2023, INE Bolivia · survey-weighted · women with children · whiskers = 95% CI

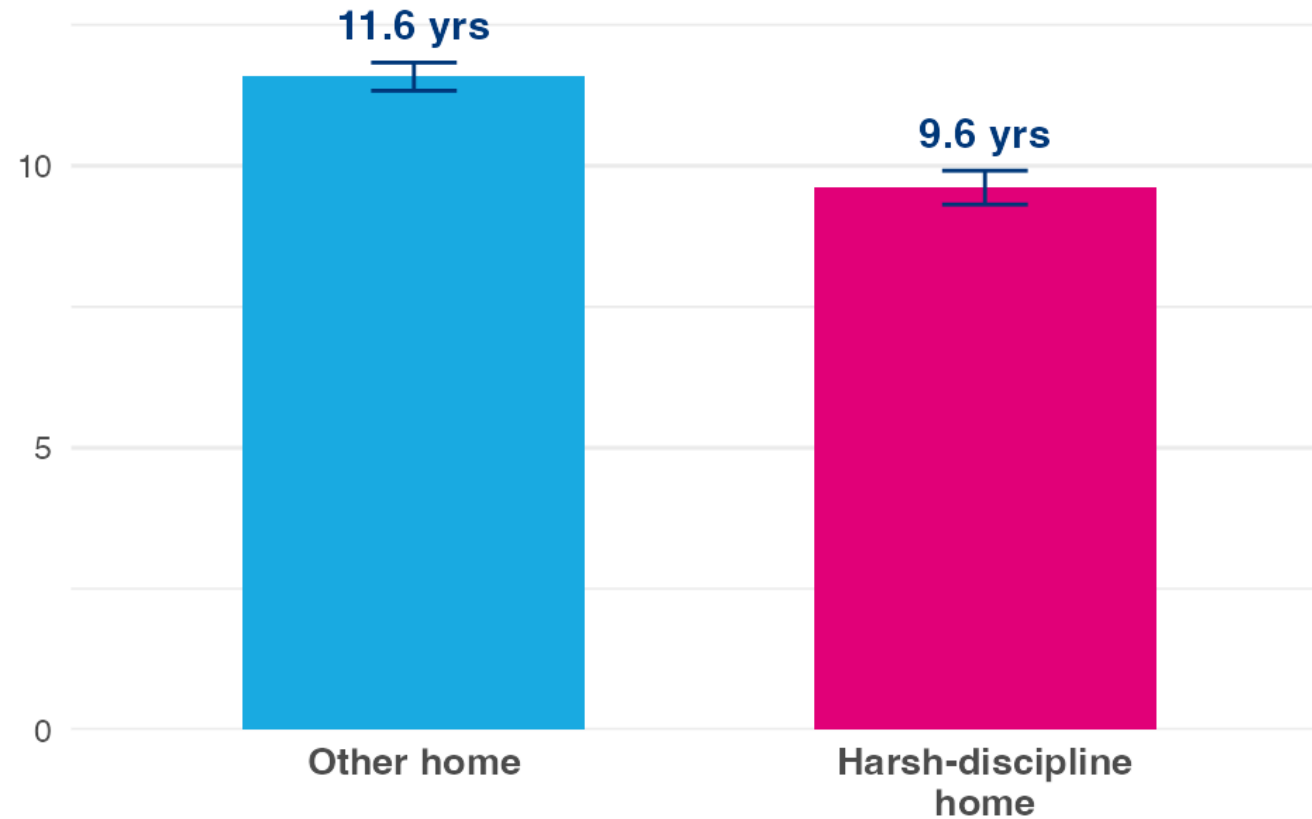
Harsh physical discipline of children (EDSA 2023, survey-weighted, women with children). Both axes raise risk **independently**: rural homes report 52.3% vs 32.2% urban (a ~20 pp gap), and indigenous homes 45.9% vs 35.0% non-indigenous (a ~11 pp gap). Rural location is the stronger axis, but indigeneity adds risk on its own — so targeting follows **rural + indigenous**, not a single dimension.

On the sex of the child: this is a **household-level** measure reported by women about discipline of children in the home; EDSA does **not** disaggregate it by the child's sex (and elsewhere physical discipline often falls more on boys). The point for this case is therefore about the **household environment** — a violence-prone home is a risk setting that girls in it **experience or witness**, and which tracks lower schooling.

Harsh discipline tracks lower schooling

Years of schooling by household discipline

Women 12–49 · EDSA 2023, survey-weighted



EDSA 2023, survey-weighted. Women in harsh-discipline homes average ~2 fewer years of schooling than women in other homes. *Note:* the violence/discipline module in EDSA is asked of **women only**, so these figures reflect **women's responses** about their own households; comparable male-respondent data are not collected. This motivates the regression that follows.

Violent discipline is associated with lower schooling

Years of schooling (women 12–49)	Association	Std. error	p
Harsh physical discipline in home	−0.90	0.16	<0.001
Indigenous	−1.10	0.20	<0.001
Rural	−0.05	0.32	0.87
Wealth Q2 vs Q1	+1.42	0.25	<0.001
Wealth Q3 vs Q1	+2.74	0.37	<0.001
Wealth Q4 vs Q1	+4.49	0.35	<0.001
Wealth Q5 vs Q1	+6.69	0.35	<0.001
Age	−0.12	0.01	<0.001

Living in a harsh-discipline household is associated with ~0.9 fewer years of schooling, holding wealth, ethnicity, area and age constant — the home-environment (violence) pathway. *Early pregnancy — an even stronger and distinct pathway — is shown on the next slide.*

Early pregnancy — a second, distinct pathway

Beyond the home-environment (violence) pathway, **adolescent pregnancy** is the **single strongest schooling correlate** in the data — a separate vulnerability the violence story does not capture.

Schooling association (women)	Estimate	n	p
Early pregnancy (< 18)	−2.9 years	391	<0.001
Harsh-discipline home (<i>for comparison</i>)	−0.9 years	4,678	<0.001

- The early-pregnancy penalty is **roughly 3× larger** than the violence-home penalty — early childbearing is strongly associated with leaving school.
- This is a **distinct pathway** calling for the **empowerment layer** (ELA-style clubs), whose evidence base includes **reduced early pregnancy and marriage** (Bandiera et al., 2018) — not the parenting/violence component.

2 · Evidence → Theory of Change

From diagnosis to three matched responses

The diagnosis points to three distinct problems, each with a different evidence-based answer:

Problem 1 — learning, not access

Enrolment is high in the early teens but **falls to 86% by 15–19** (the senior-secondary transition); the binding gaps are in **learning quality**, not access.

- Isolated hardware rarely improves learning
(Cristia et al., 2017)
- Tools work **with structured pedagogy + teacher training**
(Gómez-Ruiz & Franco, 2023)
- Gender-responsive content can lift girls' **STEM aspirations**
(Outes-León et al., 2020 — a design feature)

→ Anchor: **Evans & Yuan (2022)**, 73+ RCTs.

Problem 2 — staying in school

Retention falters for the most vulnerable girls; empowerment and safe spaces keep them enrolled.

- Girls' clubs **raise enrolment (+8.5 pp)**
(Bandiera et al., 2018 — ELA)
- Life-skills + mentoring tackle the drivers of drop-out, including early pregnancy
→ Anchor: **Bandiera ELA**, ϕ -adjusted to Bolivia.

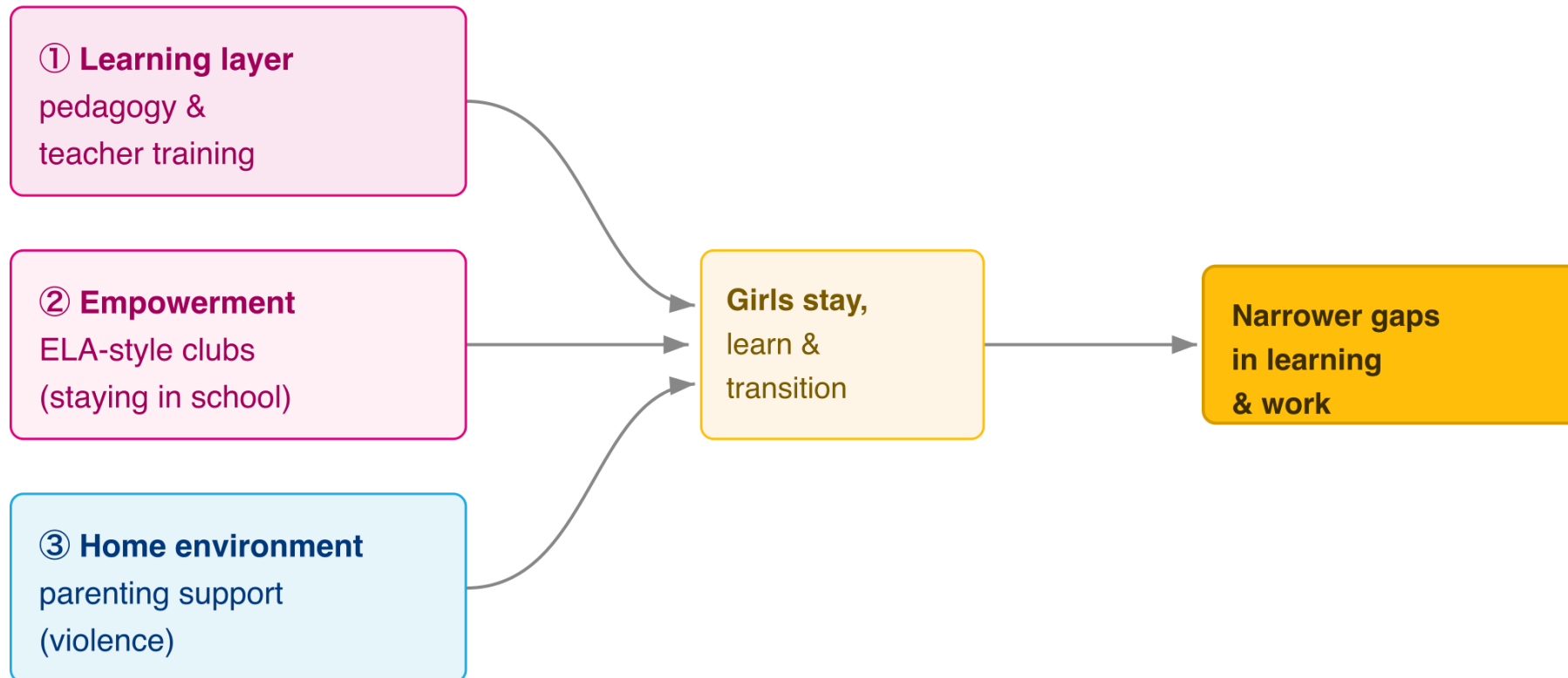
Problem 3 — the home environment

Violent discipline is concentrated among marginalised girls and tracks lower schooling.

- Parenting programmes **cut violence** against children (IRR 0.55)
(Cluver et al., 2017)
- **Low-cost smartphone delivery**
(Janowski et al., 2025 — ParentApp)
→ Anchor: **Cluver (effect) + Janowski (delivery)**.

Each problem maps to one layer: the **learning layer** (the costed BCR anchor), the **empowerment layer** (ELA — the attendance/retention answer, ϕ -adjusted), and the **home-environment layer** (parenting, for the violence pathway). Hardware is an enabler, not the intervention.

What we are building — evidence-matched layers



Each layer answers a diagnosed problem: the **learning layer** (pedagogy — the costed BCR anchor), the **empowerment layer** (ELA — the attendance/retention answer, ϕ -adjusted to Bolivia), and the **home-environment layer** (parenting, for the violence pathway, costed from the Janowski delivery model). Hardware is an enabler, not the intervention.

3 · Projections — Aula Conectada in Bolivia

Direct anchors — digital education and gender norms

Programmes with measured effects **on the outcomes** Aula Conectada seeks to improve — one per layer:

Layer	Anchor	Country	Measured effect	Used for
Learning	Evans & Yuan (2022), 73 RCTs	Meta-analysis	+0.07 SD learning (girl-specific)	Learning benefit (in SD)
Empowerment	Bandiera et al. (2018) — ELA	Sierra Leone	+8.5 pp enrolment	Attendance → earnings (ϕ -adjusted)
Home environment	Cluver et al. (2017) — PLH Teens	South Africa	IRR 0.55 on child abuse	Violence↓ → attendance (chain)

Each anchor is matched to the outcome it actually measured (Route A — no cross-unit conversion). The **learning** effect stays in SD and is reported as a learning benefit, not converted to attendance. The **ELA** and **parenting** layers carry the attendance effects (in pp), ϕ -adjusted to Bolivia. The **ELA anchor also reduced early pregnancy and marriage** — the second adolescent-girl pathway in the diagnosis — though here we monetise only its attendance effect (a conservative choice). Delivery model for the parenting layer: Janowski et al. (2025) ParentApp — low-cost, smartphone-based. Full ϕ logic and the CBA chain in the assumptions slide and appendix.

Key assumptions

The five ϕ factors are claims, not estimates

- $\phi_{\text{pop}} = 1$ when the source and target are identical in age and % rural
- $\phi_{\text{baseline}} = 1$ when Bolivia has full headroom; less when the indicator is already close to its ceiling
- ϕ_{geo} is banded by Haversine distance (1.00/0.85/0.65/0.50)
- ϕ_{econ} is capped at [0.30, 1.00] for stability
- ϕ_{delivery} reflects government implementation versus NGO/research

Robustness

- Multiplicative form is **conservative**: each $\phi \leq 1$, so the projection only attenuates the external effect, never amplifies
- The ELA anchor is the **most transported** (Sierra Leone, post-conflict, in-person); we show its BCR contribution across $\phi_{\text{ELA}} \in [0.40, 0.70]$ rather than a single point
- Sensitivity to alternative weightings in the technical methods paper (in preparation)

Synthesis — expected returns from Aula Conectada

Three layers, each anchored on the outcome its evidence **actually measured** — no cross-unit conversion.

Layer	Source effect	ϕ (transport)	Bolivia effect	Channel
Learning (digital pedagogy)	+0.07 SD (Evans & Yuan)	≈ 0.86	+0.06 SD	reported in SD — <i>not</i> converted to attendance
Empowerment (ELA-style)	+8.5 pp enrolment (Bandiera)	0.55	+4.7 pp	keeps girls in school → school-years → earnings
Home environment (parenting)	IRR 0.55 × prevalence (Cluver)	0.85	+1.9 pp	violence↓ → attendance → school-years → earnings

The ϕ (transport) column is the *composite* attenuation factor applied to each source effect ($\phi = \prod \phi_k$ over the five dimensions — population, baseline, geography, economy, delivery). The full per-factor breakdown for each anchor is in the projection output (`05_projections.R`); see the assumptions slide for the ELA ϕ range.

The two **attendance** effects act through **different mechanisms** (empowerment/safe-spaces vs the home environment), so they are **added**: ≈ 6.6 pp combined, feeding the Mincer → earnings chain. The **learning** effect is reported separately in SD — we deliberately do **not** claim a digital-only attendance effect, since a device alone does not address why girls leave school.

CBA assumptions — every number, sourced

Parameter	Value	Source
Mincer return per school-year	7.5%	EH 2024, Bolivian women (A12)
Female annual wage	\$3,500	INE EH 2024 weighted median, age 25–54
Working life · discount rate	35 yrs · 3%	LAC norm · World Bank Educ. default (2–5% sens.)
PV per attended school-year	\$3,520	project Mincer pipeline (08_bcr_estimation.R) — discounted lifetime earnings premium
pp → attended school-years	+2.4pp ⇒ +0.10 yr	over ~4 secondary years
ELA enrolment effect	+8.5 pp	Bandiera (2018), Sierra Leone
ϕ (ELA transport)	0.40–0.70	Sierra Leone→Bolivia (range shown)
Violence→attendance gap	–13.1 pp	EDSA 2023, women 12–24 — assoc. treated as causal
Cluver effect	IRR 0.55 (≈45% ↓)	Cluver (2017), PLH Teens RCT
Harsh-discipline prevalence	38%	EDSA 2023, mothers
ϕ (parenting transport)	0.85	LMIC→Bolivia
ELA cost · Parenting cost	\$26 · \$10	benchmarks (ELA \$17–35; Janowski \$8–12), not Bolivia-costed
Learning bundle cost	\$152/girl	deck costing (A7)

Parenting chain: $+1.9\text{pp} = 13.1 \times 0.45 \times 0.38 \times 0.85$. **Combined attendance** = ELA + parenting (different mechanisms, additive). The two assumptions to strengthen with a Bolivia pilot: the violence→attendance causal step, and the ELA transport factor ϕ .

4 · Costing

Unit costs — assumptions and sources

We construct **three scenarios (low / central / high)** from documented regional benchmarks. All values are annual per girl, in USD 2024.

Component	Low	Central	High	Source
Hardware amortised	30	48	60	Plan Ceibal Uruguay TCO: \$192/4yr = \$48/yr (ACM TechNews 2013); low = bulk procurement, high = harder-to-reach areas
School connectivity	12	20	30	Plan Ceibal connectivity+fibre: ~\$20/child/yr; rural premium drives high scenario
Digital content & platform	5	10	15	IDB EdTech reviews 2018-2022: licensing + curated content cost \$5-15/child/yr
Teacher training (annualised)	8	15	25	Bolivia secondary teacher salary \$716/mo (boliviainformacion.com 2024); 20 days PD over 2 years amortised across ~1000 girls per cohort
Pedagogical support and M&E	5	8	12	Plan Ceibal admin: ~\$27/yr (OLPC News 2010); fraction allocated to gender-pedagogy support
TOTAL per girl/year (USD)	60	101	142	Sum of components

Notes: (1) Bolivia teacher salaries 2024: ~Bs 3,206/mo primary (~\$462), Bs 4,947/mo secondary (~\$716); rural permanence bonuses could lower net cost in priority zones. (2) Figures exclude initial capital (electrical infrastructure, classrooms) — assumed pre-existing. (3) International cost references (e.g. Plan Ceibal ~\$75–100/child) are context for the hardware scenario only; our central case is the pedagogy bundle, not device procurement.

Component costs used in the CBA

The three layers enter the cost-benefit analysis at the following per-beneficiary costs (USD 2024):

Component	Cost	Basis
Learning bundle (pedagogy + teacher training)	\$152 / girl	deck costing, Bolivia-anchored
ELA-style empowerment	\$26 / girl	benchmark (\$17–35/yr clubs) — <i>not yet Bolivia-costed</i>
Parenting (digital, Janowski model)	\$10 / caregiver	benchmark (\$8–12) — <i>not yet Bolivia-costed</i>

The two benchmark costs are reported transparently and feed the layered cost-effectiveness; a dedicated Bolivia costing exercise would refine them before any scale-up decision.

Programme scale — girls reached

Aula Conectada as investment in **adolescent girls 10–19**, with three priority-coverage scenarios:

Scenario	N girls	% of total	Total cost (~\$188/girl)	Coverage
Pilot (8 priority departments)	~80,000	7%	~\$15 M	Rural+indigenous+poor
Intermediate scale	~280,000	24%	~\$53 M	Broad priority subset
Full scale — all priority girls ★	~568,000	48%	~\$107 M	All girls with ≥1 vulnerability
Universal (all girls 10–19)	~1,186,000	100%	~\$223 M	National coverage

★ **Recommended (central planning) scenario:** full priority scale (~568,000 girls, ~\$107 M total). **Population figures** from weighted EH 2024 (~1.19 M girls 10–19; Table A1). Total cost $\approx N \times \$188/\text{girl} = \$152 \text{ learning} + \$26 \text{ ELA} + \10 parenting (per caregiver of a reached girl); a Bolivia costing exercise would refine the ELA/parenting benchmarks. We do **not** recommend universal coverage — it would dilute impact in households without digital-access needs. The **parenting layer** reaches the *caregivers* of these girls through the same priority targeting, concentrated on the rural and indigenous households where violent discipline is most prevalent.

5 · ROI — Cost-benefit analysis

Cost-effectiveness by layer

Each layer is anchored on the outcome its evidence measured. The **learning** layer's benefit is in SD (learning); the **ELA** and **parenting** layers deliver attendance in pp.

Layer	Effect (ϕ -adj.)	Cost	Cost per unit gain
Empowerment (ELA)	+4.7 pp attendance	\$26 / girl	~\$6 per +1 pp
Parenting (violence↓)	+1.9 pp attendance	\$10 / caregiver	~\$5 per +1 pp
<i>Learning (digital pedagogy)</i>	<i>+0.06 SD learning</i>	<i>\$152 / girl</i>	<i>~\$25 per 0.01 SD</i>

ELA at $\phi=0.55$; parenting via the Cluver→EDSA chain at $\phi=0.85$. The two attendance layers are highly cost-effective *per pp* because their benchmark costs are low. The learning layer's benefit is a learning-quality gain (in SD), reported in its own unit — **~\$25 buys 0.01 SD** ($\$152 \div 6$) — so we never convert across units or double-count.

Combined BCR — shown as a range, not a point

The headline depends most on how much the ELA effect transports from Sierra Leone. We show it across ϕ rather than pick one number:

ϕ (ELA transport)	ELA pp	Parenting pp	BCR: ELA only	BCR: ELA + parenting
0.40 (conservative)	3.4	1.9	2.8	4.1
0.55 (central)	4.7	1.9	3.9	5.1
0.70 (optimistic)	5.9	1.9	4.9	6.1

What the BCR columns count: the **numerator** is the monetised *attendance* $\rightarrow$ *earnings* benefit only — “ELA only” counts the ELA attendance effect; “ELA + parenting” adds the parenting attendance effect. The **learning layer’s +0.06 SD benefit is NOT in the numerator** (it stays in SD, reported separately), but the learning **cost (\$152) IS in the denominator** of both columns alongside the ELA/parenting costs. So these BCRs already pay for all three layers while banking only the attendance benefit — i.e. a **conservative floor**, not a learning+ELA+parenting total. At the most conservative ELA transport ($\phi=0.40$), ELA alone roughly reproduces the original access-only BCR (~2.8); parenting adds ~1.3. The combined figure is **illustrative** — it rests on the violence $\rightarrow$ attendance causal assumption and the ELA ϕ ; the robust result is the **per-layer cost-effectiveness** on the previous slide.

Benefit value chain — combined attendance to earnings

How the two attendance effects become a present-value benefit per girl (central case, $\phi_{\text{ELA}}=0.55$):

Step	Calculation	Result
1 • ELA attendance	Bandiera +8.5 pp $\times \phi$ 0.55	+4.7 pp
2 • Parenting attendance	13.1 pp $\times$ 0.45 (Cluver) $\times$ 0.38 (prev.) $\times$ 0.85	+1.9 pp
3 • Combined (different mechanisms, additive)	4.7 + 1.9	+6.6 pp
4 • School-years gained	+6.6 pp over ~4 secondary years	+0.28 yr
5 • Earnings (PV)	+0.28 yr $\times$ \$3,520/school-year	~\$970
= Benefit vs. cost	\$970 $\div$ (\$152+\$26+\$10)	BCR $\approx$ 5.1

Reading the result: +6.6 pp attendance $\approx$ 0.28 of a school-year (~3–4 months) more schooling per girl. The ~\$970 is a one-off total — the *present value today* of the extra earnings that 3–4 months of schooling generates across her **whole ~35-year working life**, already discounted at 3%. It is **not** \$970 per year. Divided over the cost of all three layers (\$188), that is a BCR $\approx$ 5.1. The **learning** layer's +0.06 SD benefit is *additional* and reported in SD — not in this earnings figure — so the monetised BCR stays a floor. Violence→attendance treated as causal for the projection (flagged). All constants on the CBA assumptions slide.

Scaled to Bolivia — the per-girl gain as a national return

The **\$970 per girl** is a present value; multiplied across the girls reached, it becomes a substantial national return.

Coverage scenario	Girls reached	Lifetime earnings (PV)	Programme cost
Pilot (8 priority departments)	~80,000	~\$78 M	~\$15 M
Intermediate scale	~280,000	~\$272 M	~\$53 M
Full priority scale ★	~568,000	~\$551 M	~\$107 M
Universal (all girls 10–19)	~1,186,000	~\$1.15 B	~\$223 M

Lifetime earnings = girls reached × **\$970 PV/girl** (attendance→earnings only — the conservative floor; the learning-quality gain and the broader benefits are additional). At the recommended **full priority scale**, that is roughly **\$551 million** in discounted lifetime earnings for about **\$107 million** invested. These are present values discounted at 3%, not annual flows.

What the BCR does and does not count

The BCR monetises **only** the attendance → earnings channel. It is the **floor**, not the ceiling.

Monetised

- Attendance / retention gains
- Bolivia Mincer return (7.5%/yr)
- Lifetime labour earnings (35 yrs)

Not monetised (*documented, would only raise the return*)

- Gender norms and agency
- Household violence reduction
- Parent–child communication
- Intergenerational effects

Conclusions



The evidence points to a clear, gender-focused design: invest in **how marginalised girls learn** and in **the home environment they learn from** — not in hardware.

Who is most vulnerable

- **Rural and indigenous girls:** the largest schooling gaps (rural 8.7 vs urban 11.8 years) and the highest exposure to **violent discipline** at home (rural 52% vs 32% urban; indigenous 46% vs 35%).
- Girls in **violent-discipline homes:** ~2 fewer years of schooling (association, $p < 0.001$).
- Girls facing **early pregnancy:** the single strongest schooling correlate (−2.9 years).

What to fund

- **Learning layer** — gender-responsive pedagogy, teacher training, computational-thinking content for girls (Evans & Yuan, 73+ RCTs). Hardware is an enabler only.

What to add

- **Empowerment layer** (ELA-style) — the retention answer, addressing drop-out drivers including **early pregnancy** (the strongest schooling correlate, −2.9 yrs).
- **Parenting component** (Cluver effect + Janowski low-cost digital delivery) for the violence pathway.

Expected effect and return

- **Per-layer cost-effectiveness** (the robust headline): ELA ~\$6 per +1 pp attendance, parenting ~\$5 per +1 pp, learning ~\$25 per 0.01 SD.
- **Combined BCR ≈ 4.1 – 6.1** (ϕ -range, central ~5.1) through attendance→earnings alone — a conservative **floor**. Broader benefits (safer homes, the next generation) are **documented but not monetised**, so the true return is higher.

6 · Methodological Appendix

A0 • Cost-benefit analysis assumptions

- **Estimated costing.** Unit costs are projected from regional benchmarks (Plan Ceibal Uruguay, IDB EdTech reviews, Bolivia teacher salary scale). An official programme costing may refine the range — we recommend reviewing before scale-up decisions
- **Management and coordination costs.** Figures include administrative overhead (10-20%) but exclude in-country monitoring, inter-institutional coordination, and political/operational costs in remote areas. These could add 10-20% extra in rural scenarios
- **Benefits only via the education-labour channel.** We do not quantify benefits in agency, reproductive health, violence reduction, or domestic productivity — all potentially significant but harder to monetise defensibly
- **Mincer estimated from Bolivian data.** The return per year of schooling is estimated from EH 2024 (7.5%/year for women, 95% CI [6.6%, 8.4%]). The estimate does not incorporate the effect of schooling on the *probability* of employment — the total benefit is likely larger than projected here
- **3% discount rate** is the World Bank Education Sector standard; high sensitivity to discount rate (at 5% central BCR drops proportionally)
- **Analysis excludes positive externalities** on younger sisters, community norms, and intergenerational effects on future daughters — all documented in the literature but outside the scope of this conservative BCR

A1 · Data and sample — both surveys

EH 2024 (education, access, Mincer) — INE Bolivia · Nov–Dec 2024 · 12,718 households, 39,497 persons · analytical subsample **6,254 adolescents 10–19** · design `upm/estrato/factor`, `srvyr (lonely.psu="adjust")`. Key vars: sex `s01a_02`; age `s01a_03`; indigenous = childhood/spoken language codes 1–37, 70, 71; poverty `poor`; disability Washington Group ≥ 3 ; HH internet `s06a_19`. Weighted OLS, PSU-clustered SE (`fixest::feols`). Microdata: anda.ine.gob.bo/catalog/163.

EDSA 2023 (home environment: violence, discipline, agency) — INE Bolivia DHS round · Sep–Dec 2023 · **14,545 women 12–49**; discipline module asked of **mothers (4,678)** · design `upm/estrato/ponderadorm`, Taylor linearization (`srvyr`). Key vars: harsh discipline `ms11_1147/1148`; IPV `ms11_1105/1108`; attendance `ms01_0114`; years `aestudio`; indigenous self-ID `ms01_0108`; wealth quintile `riqueza` (Vivienda, merged by `folio`). Microdata: anda.ine.gob.bo/catalog/119.

Why two surveys: EH carries the education/earnings case; EDSA carries the home-environment diagnosis. They are independent and **not individually linkable** — so EDSA calibrates the diagnosis and the violence→attendance association, never merged into EH at the unit level. At 15–19 the two agree within ~1pp on attendance, schooling, % rural and % indigenous (slide 3), supporting joint use.

A3 · Empirical strategy

Base model (LPM with department fixed effects):

$$Y_{ihd} = \alpha + \beta_F F_i + \beta_R R_i + \beta_I I_i + \beta_D D_i + \beta_P P_i + \gamma X_{ihd} + \delta_d + \varepsilon_{ihd}$$

where F = Girl, R = Rural, I = Indigenous, D = Disability, P = Poor, X = controls (age).

Intersectional model with two-way interactions (× Girl):

$$Y_{ihd} = \alpha + \beta_F F_i + \gamma X_{ihd} + \sum_{k \in \{R, I, D, P\}} \theta_k (F_i \times k_i) + \delta_d + \varepsilon_{ihd}$$

Each θ_k measures how much larger or smaller the gender gap is in that subgroup. For example, θ_R captures whether the gender gap between rural girls and boys differs from the gap between urban girls and boys.

Intersectional model with three-way interactions (robustness):

$$Y_{ihd} = \alpha + \text{main effects} + 2\text{-way} + \theta_{F \cdot R \cdot I} (F_i \times R_i \times I_i) + \delta_d + \varepsilon_{ihd}$$

and analogous for $F \times R \times P$ and $F \times I \times P$. Three-way interactions capture whether the gender gap in a compound intersection (e.g. rural indigenous girls) differs from what the two-way interactions would suggest separately.

Estimation by weighted OLS using **factor** (expansion factor); standard errors clustered at PSU level (**fixest::feols**).

A4 · Full coefficients table — main model

Variable	HH internet	Enrolled	Attending	Any HH device	Age-for-grade delay
Girl (1=yes)	0.013	-0.001	-0.005	0.004	-0.065***
	(0.011)	(0.008)	(0.008)	(0.005)	(0.014)
Rural	-0.241***	-0.068***	-0.070***	-0.044**	0.049**
	(0.039)	(0.020)	(0.020)	(0.022)	(0.021)
Poor (INE)	-0.142***	0.012	0.008	-0.016*	0.058***
	(0.022)	(0.008)	(0.009)	(0.008)	(0.016)
Indigenous	-0.007	-0.061***	-0.063***	-0.004	0.018
	(0.036)	(0.019)	(0.019)	(0.016)	(0.024)
Disability (WG3+)	-0.012	-0.472***	-0.469***	0.039***	0.356***
	(0.069)	(0.066)	(0.066)	(0.011)	(0.052)
N	7,517	7,517	7,517	7,517	7,517
R ²	0.150	0.141	0.140	0.034	0.058
Dept FE	Yes	Yes	Yes	Yes	Yes
Weights	Yes	Yes	Yes	Yes	Yes
Cluster SE (PSU)	Yes	Yes	Yes	Yes	Yes

Coefficients in decimal points (× 100 for pp). Robust SE in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

A5 • Full intersectional table

Interaction	Attending	Enrolled	HH internet	Any HH device
Girl × Rural	-0.047***	-0.043**	0.024	-0.008
	(0.018)	(0.018)	(0.030)	(0.013)
Girl × Indigenous	-0.056**	-0.052**	0.013	-0.006
	(0.022)	(0.023)	(0.037)	(0.013)
Girl × Disability	-0.032	-0.037	-0.050	-0.014
	(0.137)	(0.137)	(0.132)	(0.019)
Girl × Poor	-0.026	-0.020	0.025	-0.013
	(0.016)	(0.016)	(0.028)	(0.012)
N (each model)	7,517	7,517	7,517	7,517

Each row is the coefficient on the (Girl × subgroup) interaction in a separate regression. Coefficients in decimal points. Standard errors in parentheses. Robust SE clustered at PSU.

A7 · Formal foundation of the transportability approach

We assume that the mismatch between the source population S (the anchor) and the target T (Bolivia) decomposes into K observable dimensions, with multiplicative attenuation:

$$\hat{\beta}_T \approx \hat{\beta}_S \cdot \prod_{k=1}^K \phi_k(S, T), \quad \phi_k \in [0, 1]$$

Glossary of ϕ_k factors:

- ϕ_{pop} — standardised age and % rural
- ϕ_{baseline} — outcome headroom
- ϕ_{geo} — Haversine distance from La Paz
- ϕ_{econ} — GDP per capita ratio
- ϕ_{delivery} — 1.0 (government) or 0.7 (NGO/research)

Assumptions:

- (A1) **Sufficient observables**: the 5 dimensions capture the relevant effect modifiers
- (A2) **Separability**: effects are approximately multiplicative
- (A3) **Monotonicity**: mismatch only attenuates, never amplifies
- (A4) **SE independence**: anchor variance propagates without perturbing the ϕ_k

A8 · Bolivia-adjusted projection methodology

Algorithm for each anchor:

1. Compute ϕ_{pop} as $1 - \sqrt{d_{\text{age}}^2, d_{\text{rural}}^2}$, where d are standardised differences between source and Bolivia
2. Compute $\phi_{\text{baseline}} = (Y_{\text{max}} - Y_{\text{Bol}})/(Y_{\text{max}} - Y_{\text{anchor}})$ when measurable (otherwise 1)
3. Compute ϕ_{geo} in Haversine bands from La Paz: $\leq 1,000$ km $\rightarrow$ 1.00; 1,000–5,000 km $\rightarrow$ 0.85; 5,000–10,000 km $\rightarrow$ 0.65; $> 10,000$ km $\rightarrow$ 0.50
4. Compute ϕ_{econ} as the Bolivia/anchor GDP per capita ratio, capped at [0.30, 1.00]
5. Compute ϕ_{delivery} : 1.0 if government implementer; 0.7 if NGO/research (Vivalt 2020)
6. Multiply: $\hat{\beta}_T = \hat{\beta}_S \cdot \prod \phi_k$
7. Propagate SE: $\text{SE}(\hat{\beta}_T) = \text{SE}(\hat{\beta}_S) \cdot \prod \phi_k$
8. 95% CI: $\hat{\beta}_T \pm 1.96 \cdot \text{SE}(\hat{\beta}_T)$

Exclusion rules — age overlap <25% with 10–19, or <2 demographic variables comparable $\rightarrow$ anchor not projected.

Uncertainty — 95% CI propagated from the anchor SE. No synthetic noise on ϕ factors.

A9 • Consolidated evidence — anchors in use

Programme	Country	Age	Effect	Implementer	ToC arrow
Girls' education meta-analysis (Evans &...	LMIC (pooled)	10-19	0.070 (0.025) SD	Government	Digital education
Growth Mindset (Outes & Sanchez, full)	Perú	12-14	0.054 (0.030) SD	Government	Digital education
Growth Mindset (Outes & Sanchez, regional)	Perú	12-14	0.135 (0.051) SD	Government	Digital education
Growth Mindset (Porter et al.)	Sudáfrica	13-16	-0.040 (0.100) SD	NGO/Research	Digital education
ELA (Bandiera et al., 12-17 cohort)	Sierra Leona	12-17	7.250 (2.700) score_pts	NGO/Research	Gender norms
Irûmi — pensamiento computacional (Näsl...	Paraguay	7-8	0.089 (0.052) SD	Government	Digital education
Plan Ceibal SOLO HARDWARE (De Melo)	Uruguay	8-12	0.000 (0.050) SD	Government	Digital education
Plan Ceibal + programación (Gómez-Ruiz)	Uruguay	14-17	0.130 (0.050) SD	Government	Digital education
ProJoven (Ñopo et al.)	Perú	16-25	0.152 (0.060) ppts	Government	Labour market
ParentApp Tanzania (Janowski 2025)	Tanzania	0-17	0.130 (0.060) SD	NGO/Research	Parenting
Parenting for Lifelong Health (Cluver e...	Sudáfrica	10-17	0.100 (0.050) ppts	NGO/Research	Parenting

Complete list of external anchors used in the Bolivia projection (see [05_projections.R](#) and [R/anchors_demographics.R](#)). The full systematic review of 52 papers is in [data/lit_review_papers.csv](#). The **ToC arrow** column indicates which arc of the Theory of Change each anchor feeds.

A10 • Robustness — three-way interactions (OLS)

To test whether vulnerabilities compound, we estimate three-way interactions with OLS. If belonging to several vulnerable groups at once had a super-additive effect, the three-way coefficients would be significant for the outcomes with greatest heterogeneity.

Three-way interaction	Attending school	Enrolled	HH internet
Girl × Rural × Indigenous	-0.061 (0.045)	-0.052 (0.046)	0.050 (0.073)
Girl × Rural × Poor	-0.003 (0.036)	-0.008 (0.038)	-0.096 (0.069)
Girl × Indigenous × Poor	-0.002 (0.050)	0.005 (0.051)	0.096 (0.077)

How to read: each cell shows the three-way interaction coefficient and its standard error in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Where no asterisks appear, the coefficient is not statistically distinguishable from zero at conventional levels. **Interpretation:** in this sample, the three-way interactions are not significant, indicating that the compound effect of belonging to multiple vulnerable groups (e.g. rural indigenous girl) is not substantially larger than the sum of effects of each group separately. Gaps accumulate roughly additively. This suggests well-targeted interventions on any axis (rural, indigenous, poverty) will reach girls who fit those characteristics.

A12 · Bolivia Mincer estimation

Spec	Term	Coef	SE
A	aestudio	0.0559***	(0.0034)
A	exp_yrs	0.0376***	(0.0031)
A	exp_sq	-0.0007***	(0.0001)
A	female	-0.3257***	(0.0177)
A	rural	-0.4146***	(0.0484)
A	indigenous	-0.0101	(0.0227)
C	aestudio	0.0747***	(0.0046)
C	exp_yrs	0.0325***	(0.0041)
C	exp_sq	-0.0004***	(0.0001)
C	rural	-0.3814***	(0.0700)
C	indigenous	-0.0218	(0.0313)

Headline (Spec C, female-only): each additional year of schooling raises expected log monthly labour income by 7.5% [95% CI: 6.6% – 8.4%]. This enters the BCR directly as the Mincer return.

Sample: 13,009 adults 25–60 with positive labour income (5,327 women · 2,488 rural). Weighted OLS, PSU-clustered SE. Model: $\log(\text{ylab}) \sim \text{years_edu} + \text{exp} + \text{exp}^2 + \text{female} + \text{rural} + \text{indigenous}$.

Caveats: (i) OLS on the employed sample suffers from selection bias — the return may be over-estimated. (ii) We do not capture schooling’s effect on the *probability* of employment — this under-estimates total benefit.

Gender pay gap: the *female* coefficient (Spec A) is the conditional gender wage gap — Bolivian women earn less than men with the *same* schooling and experience.

A13 · Education outcomes by age band — overall

Early (10-14) vs late (15-19) adolescence

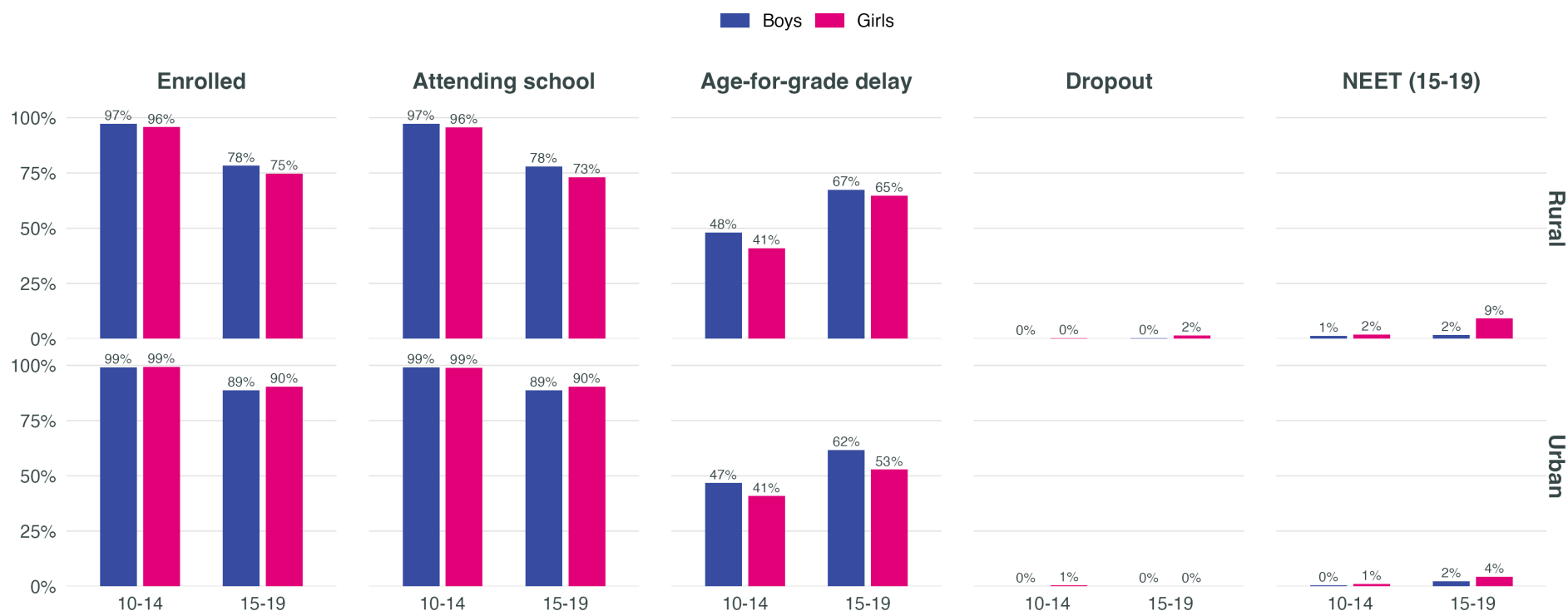


Source: INE Bolivia – Household Survey 2024. Weighted estimates.

NEET is only informative for 15-19 (in 10-14 nearly everyone is enrolled by construction).

A14 · Education outcomes by age band — area of residence

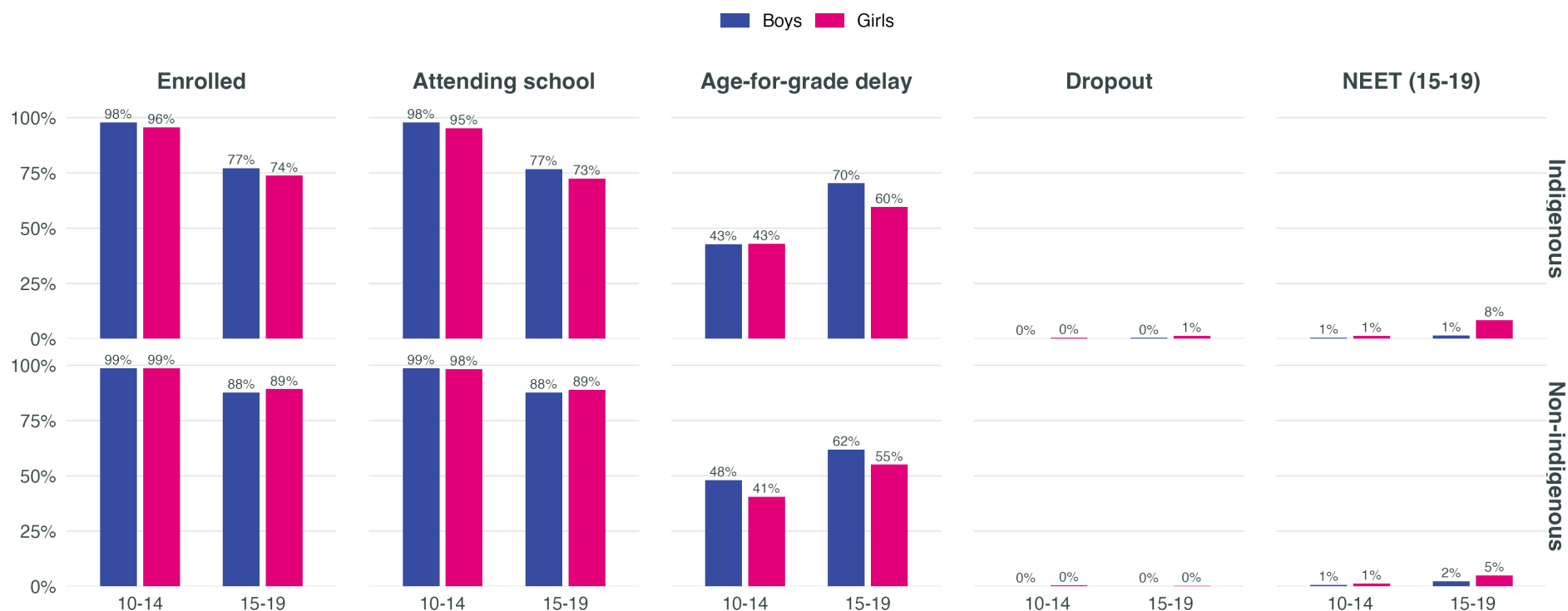
By sex, age (10-14 / 15-19) and area of residence



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

A15 · Education outcomes by age band — indigenous status

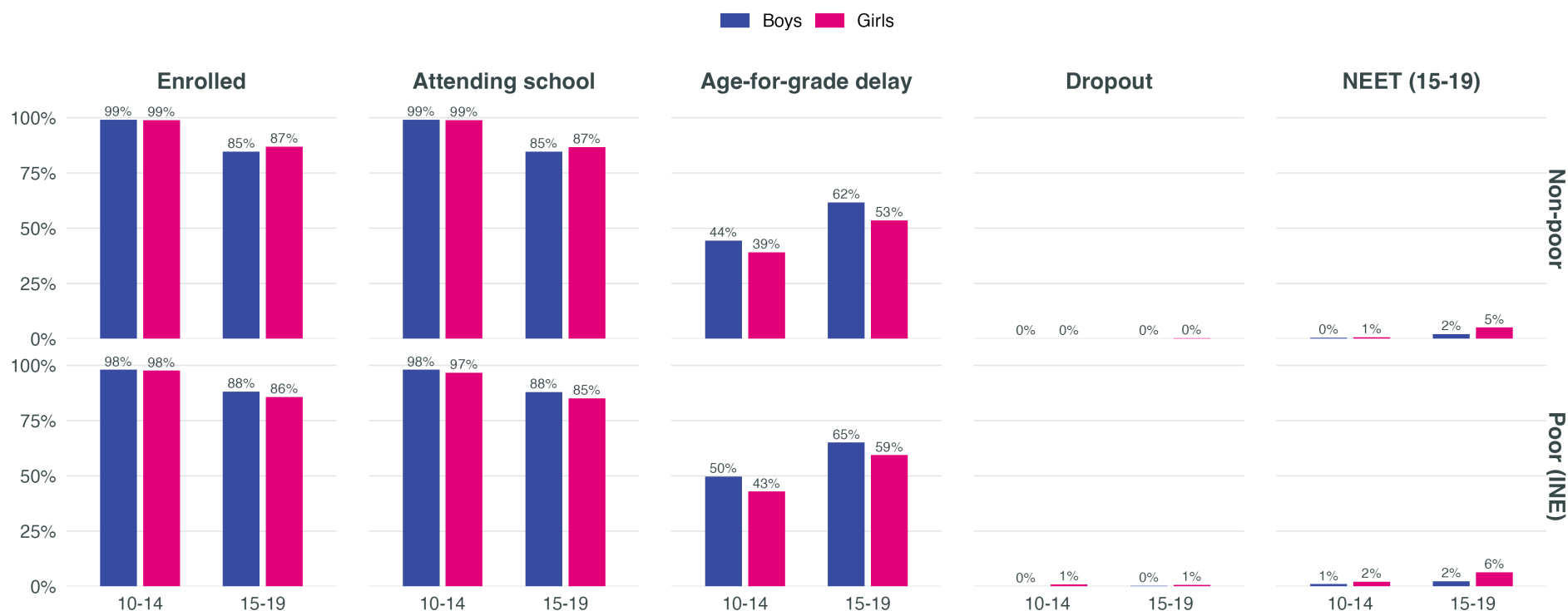
By sex, age (10-14 / 15-19) and indigenous status



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

A16 · Education outcomes by age band — poverty status

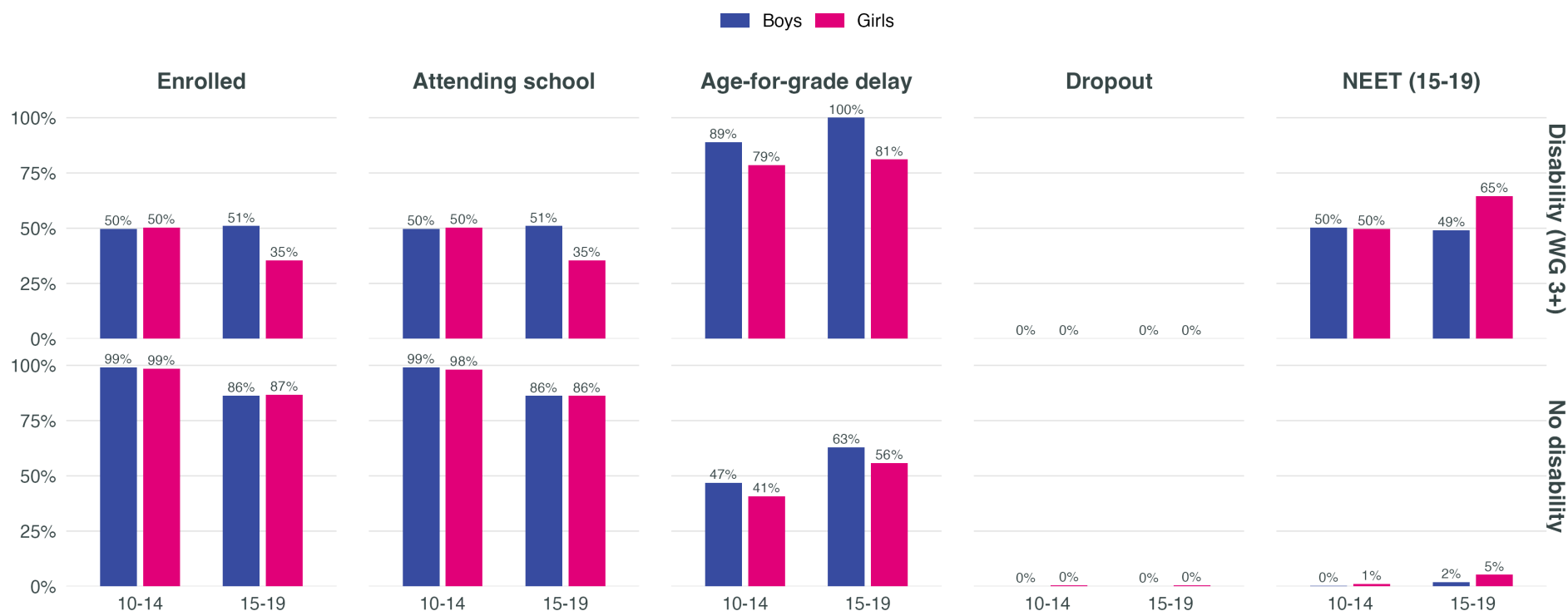
By sex, age (10-14 / 15-19) and poverty status



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

A17 · Education outcomes by age band — disability

By sex, age (10-14 / 15-19) and disability



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

Disability subsample is small (~0.4%) — interpret with caution.

A18 · Gender gaps by adolescent stage

Early (10-14) vs late (15-19) adolescence

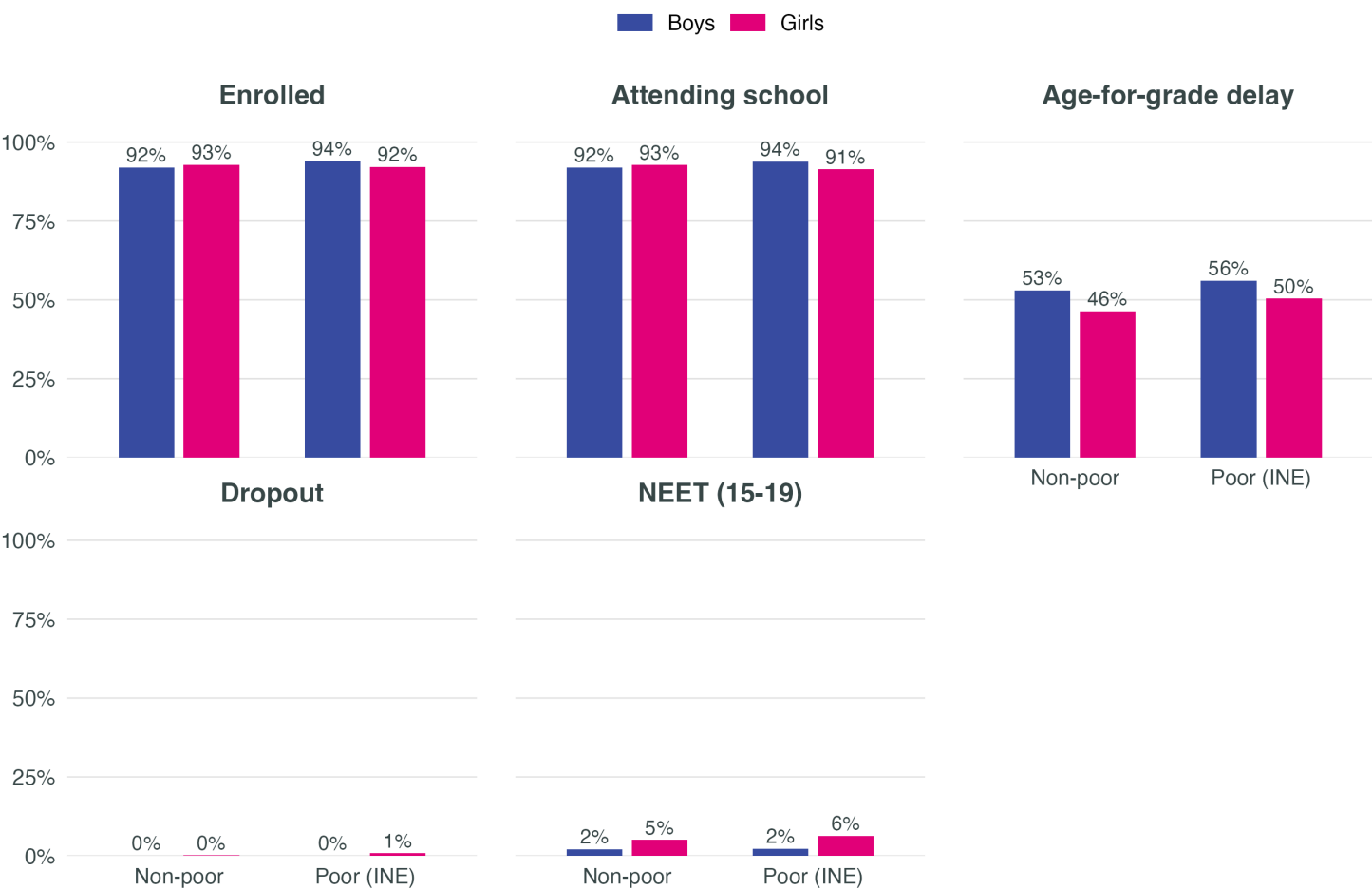


Source: INE Bolivia – Household Survey 2024. Weighted estimates.

NEET is only informative for 15-19 (in 10-14 most are in school by construction).

A19 · Education outcomes by poverty status

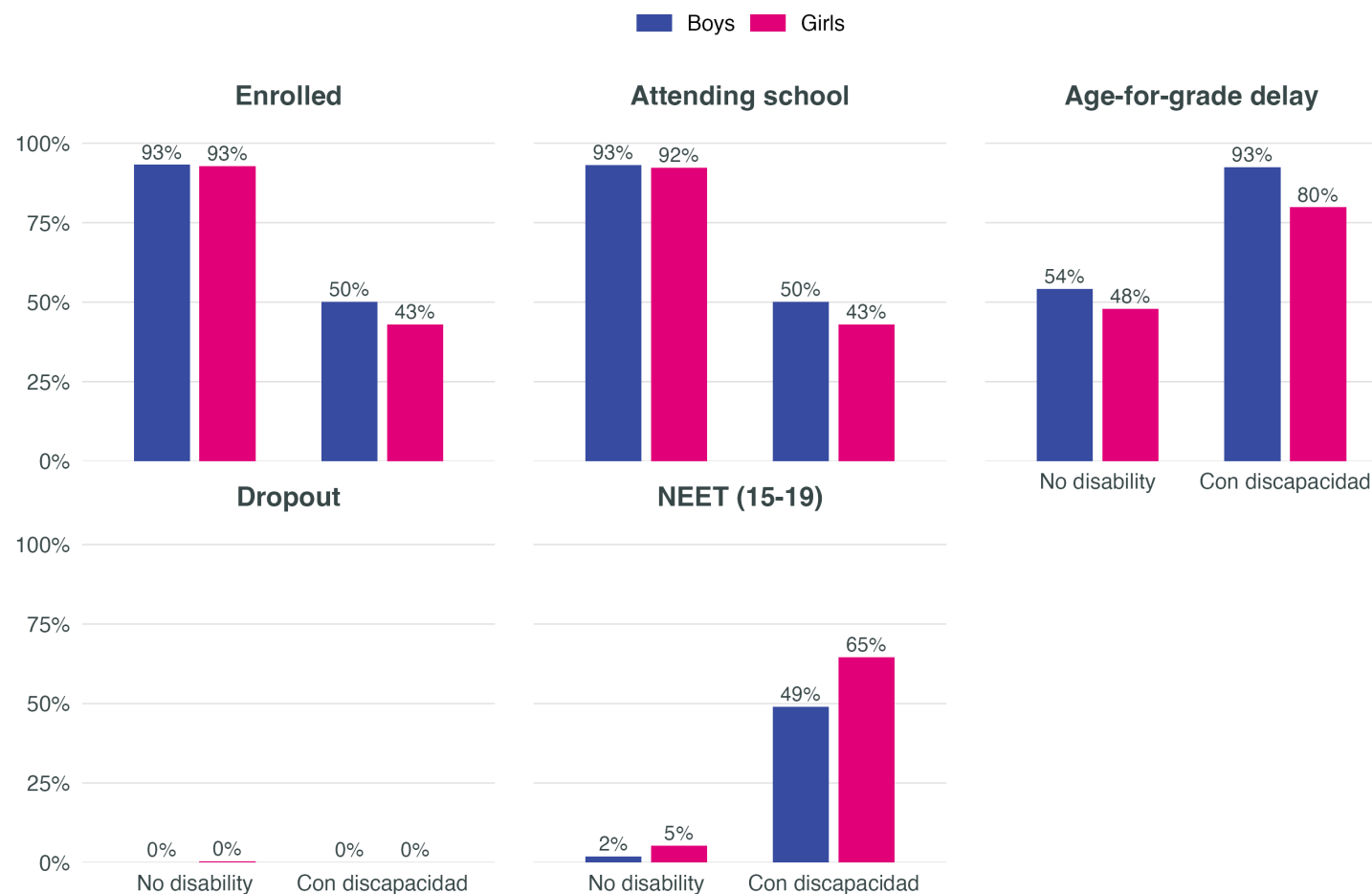
Adolescents 10–19, by sex and poverty status (INE)



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

A20 · Education outcomes by disability

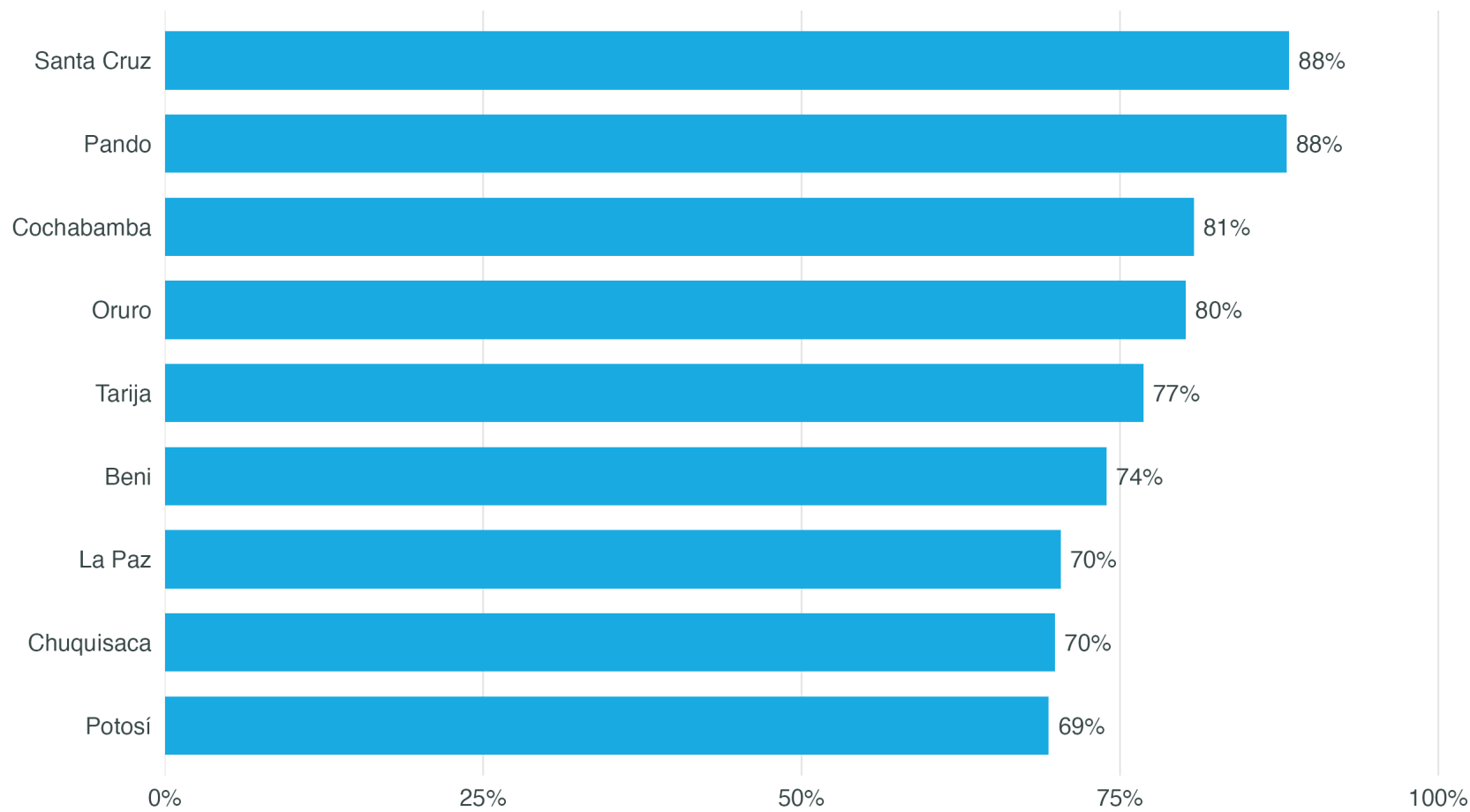
Adolescents 10–19, by sex · Washington Group severity ≥ 3



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. Small sample size (~0.4%) — interpret with caution.

A21 · Digital divide — departmental variation

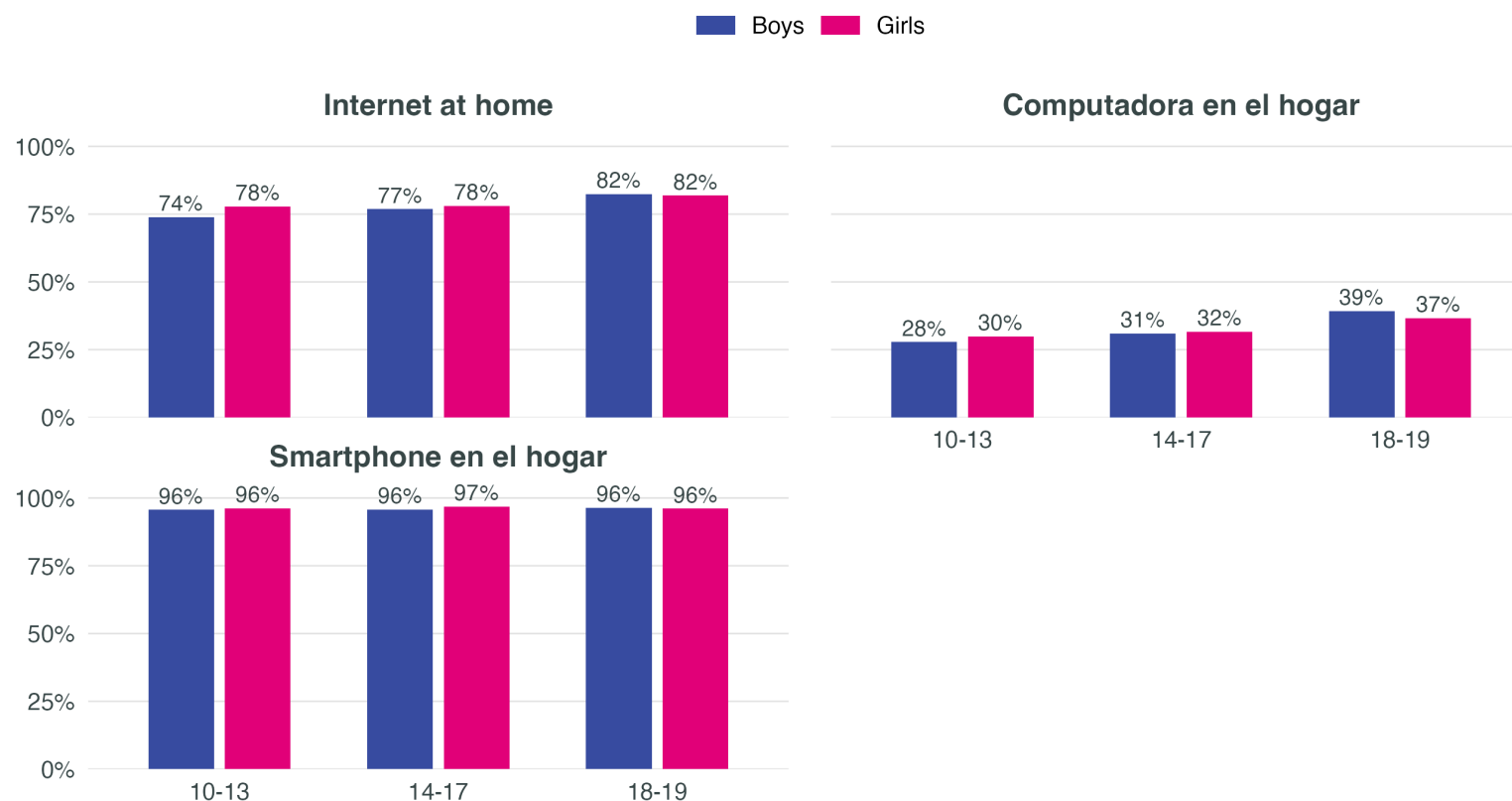
Household internet, adolescent girls 10–19, by department



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

A22 · Household digital access — no differences by sex

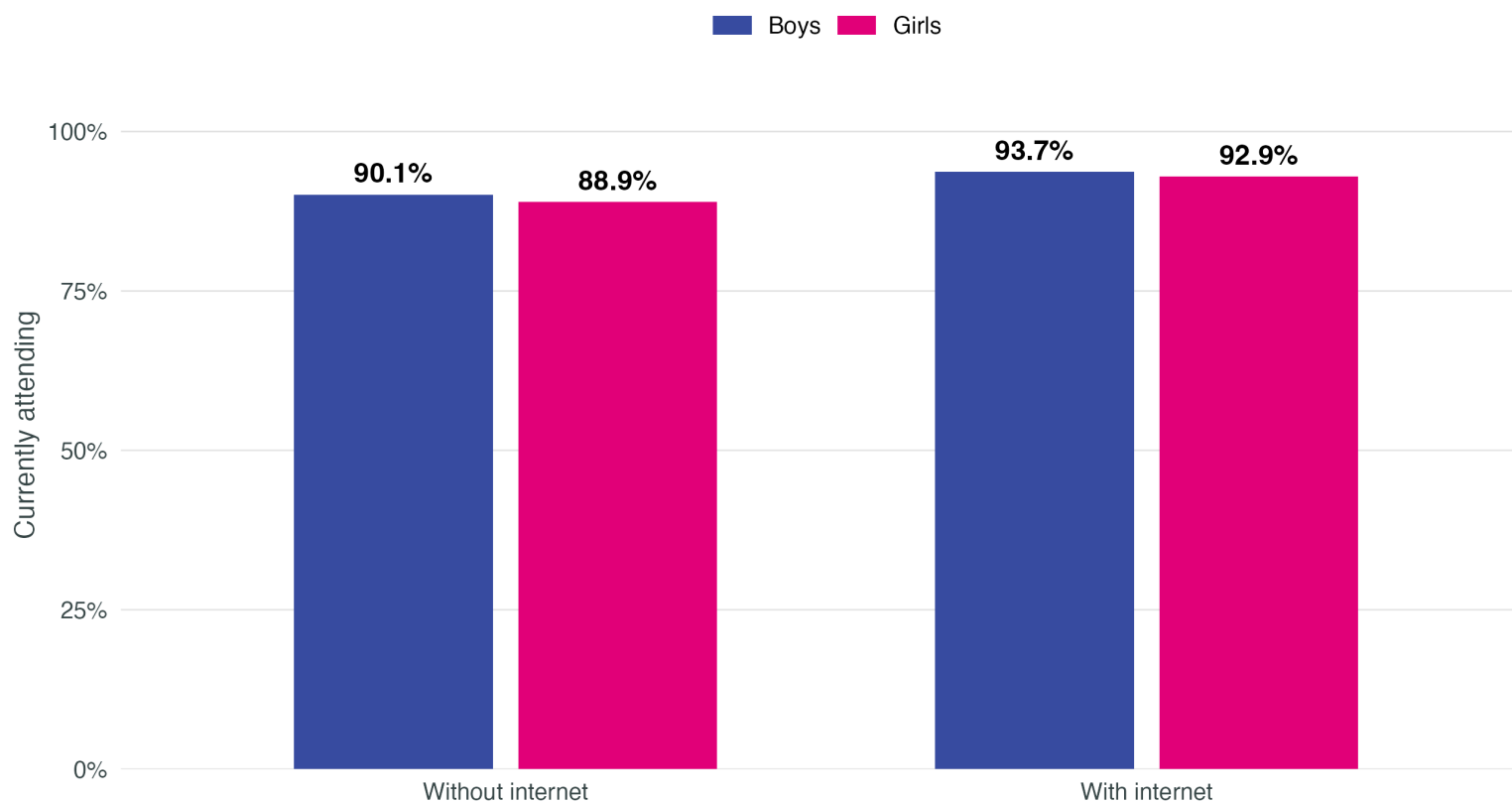
Share of girls and boys in households with each digital asset, by age group



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

A23 · Connected households and girls' attendance

School attendance rate of adolescents 10–19, by household connectivity



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Note: correlation, not causation. Households with internet also tend to have higher income, urban location, and parental education.